GREENHOUSE RISK PREDICTION SYSTEM

GRiPS

For AuTOMATO: IoT-Based Automated Greenhouse for Cherry Tomato Cultivation

Polytechnic University of the Philippines, Lopez Campus

1. WHAT IS GRiPS?

- A lightweight predictive risk system that forecasts environmental problems 15 to 60 minutes before they occur.
- Uses simple linear trend math on existing sensor data: current value + trend rate × time horizon.
- Requires zero new hardware — runs entirely on existing ESP32, sensors, relay module, Firebase, and Flutter app.
- Adds roughly 1 KB of RAM usage for a 30-minute circular buffer and ~500 lines of new code.
- Enables auto-prevention: triggers cooling, circulation, or irrigation before thresholds are breached.
- Predicts six risk domains: thermal runaway, humidity trap, dry-out, light crash, nutrient drift, and system stall.

2. WHY GRiPS IS IMPORTANT

- Lopez, Quezon has persistently high temperatures (24°C–32°C) and elevated humidity that make cherry tomato cultivation difficult.
- Early intervention prevents heat stress, flower drop, and fruit cracking that ruin entire fruit set cycles.
- Cherry tomato pollen viability drops above 29.4°C — brief heat spikes during flowering can destroy yields.
- Smallholder farmers cannot afford crop losses at ±350 per kilogram market price.
- A single prevented heat event during flowering pays for the entire system's preventive value.
- Manual monitoring is labor-intensive; farmers cannot watch thermometers continuously.
- GRiPS watches continuously and acts when farmers are absent, working, or sleeping.
- Fits the capstone scope with ~2 weeks of development and ~70% code reuse.
- Differentiates AuTOMATO from basic IoT monitors that only alert after damage occurs.
- Builds farmer trust in automation by preventing problems before they are visible.
- Reduces energy waste by using brief pre-cooling instead of extended emergency cooling.
- Maintains operation during brief internet outages using the local ESP32 buffer.
- Fully explainable: farmers understand that temperature rising 0.6°C in 10 min !’ 28.6°C in 30 min.

3. EXAMPLES

Example 1: Thermal Runaway Prevention

Scenario: Sunny afternoon in Lopez, greenhouse temperature rising.

TIMELINE

```
1:45 PM  Temperature 25.8°C, stable
2:00 PM  Temperature 26.2°C, trend +0.4°C per 10 min
2:10 PM  Temperature 26.8°C, trend +0.6°C per 10 min
```

GRiPS PREDICTION

Predicted temp in 30 min: 26.8 + (0.06 × 30) = 28.6°C

Threshold: 28.0°C | Minutes to breach: ~20 min | Confidence: 82%

WITH GRiPS (Auto-Prevention)

```
2:10 PM   GRiPS sends auto-prevention command to exhaust fan
2:15 PM   Exhaust fan running, circulation fans activated
2:30 PM   Temperature peaks at 27.4°C, never breaches threshold
2:35 PM   Farmer notified: "Heat stress prevented. Exhaust activated early."
```

Result: Temperature controlled at 27.4°C. No threshold breach.

WITHOUT GRiPS (Reactive)

```
2:30 PM   Temperature hits 28.5°C, threshold breached
2:30 PM   Reactive alert sent to farmer
2:32 PM   Farmer opens app, manually activates fan
2:35 PM   Fan begins cooling
2:45 PM   Temperature finally drops below 28°C
```

Result: 15 minutes of heat stress, potential flower damage.

Example 2: Humidity Trap Prevention

Scenario: Overcast morning, poor air circulation, humidity rising.

TIMELINE

```
8:00 AM   Relative humidity 62%, circulation fans on auto
8:30 AM   Relative humidity 68%, fans cycled off
9:00 AM   Relative humidity 73%, trend +1.7% per 10 min
```

GRiPS PREDICTION

Predicted humidity in 30 min: $73 + (0.17 \times 30) = 78.1\%$

Threshold: 75% | Minutes to breach: ~12 min | Confidence: 71%

WITH GRiPS (Auto-Prevention)

```
9:00 AM   GRiPS activates circulation fans preemptively
9:15 AM   Humidity peaks at 74%, never reaches 75%
```

Result: Disease-favorable conditions avoided.

WITHOUT GRiPS (Reactive)

```
9:12 AM   Humidity hits 75%, threshold breached
9:12 AM   Reactive alert sent, fans activate
9:30 AM   Humidity finally drops after 18 min in disease zone
```

Result: Extended period favorable for leaf mold and gray mold development.

Example 3: Dry-Out Prevention

Scenario: Hot afternoon, increased plant transpiration, substrate moisture declining.

TIMELINE

```
10:00 AM  Substrate moisture 78%, stable
11:00 AM  Substrate moisture 72%, trend -1.0% per 10 min
12:00 PM  Substrate moisture 66%, trend -1.2% per 10 min
```

GRiPS PREDICTION

Predicted moisture in 60 min: $66 + (-0.12 \times 60) = 58.8\%$

Threshold: 60% | Minutes to breach: ~50 min | Confidence: 75%

WITH GRiPS (Auto-Prevention)

```
12:00 PM  GRiPS schedules pre-irrigation cycle
12:45 PM  Irrigation pump activates for 2-minute cycle
1:00 PM   Moisture stabilizes at 63%
```

Result: Root zone never enters stress zone. Moisture stabilized at 63%.

WITHOUT GRiPS (Reactive)

```
12:50 PM  Moisture drops to 59%, threshold breached
12:50 PM  Reactive alert sent, pump activates
12:52 PM  Pump begins 2-minute cycle
12:54 PM  Moisture begins recovery
```

Result: 4 minutes of root stress, potential blossom end rot risk.

Example 4: Nutrient Drift Alert

Scenario: pH gradually shifting due to plant uptake and evaporation.

TIMELINE

```
Day 1  pH 6.2, stable
Day 3  pH 6.0, trend -0.07 per day
Day 5  pH 5.9, trend -0.08 per day
```

GRiPS PREDICTION

Predicted pH in 48 hours: 5.9 + (-0.008 × 48) = 5.52

Threshold: 5.5 | Hours to breach: ~50 hrs | Confidence: 68%

WITH GRiPS (Auto-Prevention)

```
Day 5  GRiPS alert: "pH approaching lower limit. Check nutrient solution within 2
days."
Day 6  Farmer inspects and adjusts nutrient mix
Day 7  pH stabilized at 6.1
```

Result: Preventive maintenance completed. No crop damage.

WITHOUT GRiPS (Reactive)

```
Day 8  pH drops to 5.4, threshold breached
Day 8  Alert sent, but nutrient lockout may already affect plant health
Day 9  Farmer adjusts solution
```

Result: 1–2 days of suboptimal nutrient uptake, potential deficiency symptoms.

Example 5: System Stall Detection

Scenario: Pocket WiFi intermittent connectivity, ESP32 struggling to sync.

TIMELINE

```
Normal   Heartbeat every 5 seconds
3:00 PM  Last heartbeat 8 seconds ago
3:05 PM  Last heartbeat 15 seconds ago
3:10 PM  Last heartbeat 25 seconds ago
```

GRiPS PREDICTION

Sync latency trend: +5 seconds per 5 minutes

Predicted complete disconnect: ~15 min | Confidence: 55%

WITH GRiPS (Auto-Prevention)

```
3:10 PM  GRiPS switches ESP32 to local automation mode
3:10 PM  Firebase sync marked as degraded
3:15 PM  Connectivity lost, but local automation continues
3:30 PM  Connection restored, GRiPS syncs missed data from SD card
```

Result: No automation gap. No crop exposure to uncontrolled conditions.

WITHOUT GRiPS (Reactive)

```
3:15 PM  Complete disconnect, no cloud visibility
3:15–3:45 PM  30-minute blackout period
If local automation fails: potential uncontrolled heat or humidity spike
```

Result: Unknown system status, farmer anxiety, possible crop damage.

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4. SUMMARY

GRiPS transforms AuTOMATO from a reactive monitor into a proactive guardian.

Using only simple math on existing data, it sees problems coming and acts before they arrive. For cherry tomato farmers in Lopez, Quezon, this means healthier plants, stable yields, lower energy costs, and confidence that their greenhouse is protected even when they are not watching.

KEY BENEFITS AT A GLANCE

Zero new hardware required
Runs on existing ESP32, sensors, and infrastructure

15–60 minute early warning
Forecasts problems before they become crop damage

Six risk domains covered
Thermal, humidity, dry-out, light, nutrient, and connectivity

Auto-prevention capable
Acts without farmer intervention when thresholds approach

Fully explainable math
Simple linear trends farmers can understand and trust

Energy efficient
Brief pre-cooling vs. extended emergency cooling saves power